

Asadullah Bin Rahman



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asadullahgalib007.github.io



in/asadullah007



Google Scholar

Research Interests: Image Processing, Machine Learning, Quantum Computing

Employment

- 07/25 – Present ♦ **Lecturer, Computer Science and Engineering**
World University of Bangladesh, Dhaka, Bangladesh
- 07/23 – 07/25 ♦ **Research Assistant**
IoThink Lab, Department of Computer Science and Engineering, HSTU
- 03/23 – 09/23 ♦ **Lecturer, Computer Science and Engineering**
Govt. Shahid Akbar Ali Science and Technology College (Affiliated to HSTU), Bangladesh

Education

- 07/23 – 07/25 ♦ **MSc in Computer Science and Engineering**, Hajee Mohammad Danesh Science and Technology University (HSTU), Bangladesh
CGPA 🔗 : 3.547/4.00
- 01/17 – 12/22 ♦ **BSc (Engineering) in Computer Science and Engineering**, Hajee Mohammad Danesh Science and Technology University (HSTU), Bangladesh
CGPA 🔗 : 3.31/4.00

Skills

- Quantum Computing ♦ Proficient in Qiskit, Cirq, PennyLane, Classiq; Expertise in Quantum Algorithms, Quantum Machine Learning, Quantum Key Distribution
- Machine Learning ♦ Proficient in NumPy, Pandas, SciPy, Matplotlib, OpenCV, PyTorch, TensorFlow; Expertise in Image Processing, Computer Vision
- Programming ♦ C++, Java, Python; Data Structures, Algorithms, Databases; Solved 300+ problems (CodeForces, HackerRank, UVA)
- Languages ♦ Bangla (Native), English (IELTS Academic: **6.5**, CEFR Level: B2 🔗)
- Others ♦ Linux, Git, LaTeX

Research Publications

- A. B. Rahman**, M. I. Afjal, and M. A. A. Mamun, *Deep Learning Architectures for Medical Image Denoising: A Comparative Study of CNN-DAE, CADTra, and DCMIEDNet*, 2025. arXiv: [2508.17223](https://arxiv.org/abs/2508.17223) [eess.IV]. 🔗 URL: <https://arxiv.org/abs/2508.17223>
- A. B. Rahman**, M. I. Afjal, and M. A. A. Mamun, *Systematic Evaluation of Wavelet-Based Denoising for Mri Brain Images: Optimal Configurations and Performance Benchmarks*, 2025. arXiv: [2508.15011](https://arxiv.org/abs/2508.15011) [eess.IV]. 🔗 URL: <https://arxiv.org/abs/2508.15011>
- A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, “Mitigating Noise from Biomedical Images Using Wavelet Transform Techniques,” in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2025, pp. 1–6. 🔗 DOI: [10.1109/ECCE64574.2025.11013062](https://doi.org/10.1109/ECCE64574.2025.11013062)
- A. B. Rahman**, M. Touhid Islam, M. R. Islam, M. Sohrawordi, and M. N. Sultan, “Enhanced Brain Tumor Classification from MRI Images Using Deep Learning Model,” in *2023 26th International Conference on Computer and Information Technology (ICCIT)*, 2023, pp. 1–6. 🔗 DOI: [10.1109/ICCIT60459.2023.10441064](https://doi.org/10.1109/ICCIT60459.2023.10441064)

Hyperlinks are embedded in this CV. Please click anywhere with 🔗 for details!

Teaching

- Summar 2026
 - ◇ Course Teacher, WUB. Compiler Design.
 - ◇ Course Teacher, WUB. Object Oriented Programming.
 - ◇ Course Teacher, WUB. Introduction to Programming.
- Spring 2026
 - ◇ Course Teacher, WUB. Object Oriented Programming.
 - ◇ Course Teacher, WUB. Introduction to Programming.
- Fall 2025
 - ◇ Course Teacher, WUB. Computer Fundamentals.
 - ◇ Course Teacher, WUB. Object Oriented Programming.
 - ◇ Course Teacher, WUB. Computer and Cyber Security.
- 03/23 – 09/23
 - ◇ Instructor, SASTC. Theory of Computation and Concrete Mathematics.
 - ◇ Instructor, SASTC. Computer Graphics and Image Processing.
 - ◇ Instructor, SASTC. Machine Learning and Pattern Recognition.

Research and Projects

- 02/26
 - ◇ **Multilayered Security** 🔗: Implemented and simulated a research project on Multi-Layered Security System: Integrating Quantum Key Distribution with Classical Cryptography to Enhance Steganographic Security.
- 11/24
 - ◇ **Guava Fruit Disease Dataset @ IoThink Lab** 🔗: Collaborated on dataset collection for interdisciplinary research.
- 08/24
 - ◇ **Quantum Variational Classifier @ Womanium Program** 🔗: Implemented a quantum classifier for Penguin Species Classification.
 - ◇ **Quantvolutional Neural Networks @ Womanium Program** 🔗: Developed a hybrid quantum convolutional model for MNIST Digit Classification.
 - ◇ **Quantum Regression Model @ Womanium Program** 🔗: Implemented a Quantum Machine Learning Model to learn and predict the sine function on the interval $[0, 2\pi]$.
 - ◇ **Quantum Machine Learning for Anomaly Detection @ Womanium Program** 🔗: Developed a hybrid model for anomaly detection, leading to project finalist recognition.

Professional Development

- 09/24 – 12/24
 - ◇ **QClass24/25 Fall Semester** 🔗: Achieved 96% in 3 ECTS graduate-level program on Quantum Algorithms and QKD.
- 06/24 – 08/24
 - ◇ **Womanium Quantum + AI 2024** 🔗: Coursework on Quantum Computing and AI.
- 08/24
 - ◇ **QGSS 2024 – Quantum Excellence** 🔗 🔗: Intensive quantum computing boot camp by IBM Qiskit.
 - ◇ **Classiq Diploma** 🔗: Advanced Quantum Algorithm design.
 - ◇ **Pennylane Diploma** 🔗: Quantum Machine Learning Challenge completion.
- 07/24
 - ◇ **QNickel Diploma** 🔗: Quantum Algorithms workshop.
 - ◇ **QBronze Diploma** 🔗: Introductory Quantum Computing and Programming.
- 09/20
 - ◇ **Python for Everybody Specialization** 🔗: Coursera specialization on Python fundamentals and data structures.

Awards and Achievements

- 06/25
 - ◇ **Unitary Hack 2025** 🔗 🔗 🔗: For open source contributions to the quantum software ecosystem.
- 04/25
 - ◇ **YQuantum 2025** 🔗 🔗: Solved BlueQubit's Peaked Circuits Challenge.
- 02/25
 - ◇ **MIT iQuHACK 2025** 🔗: Solved IONQ's Max Cut problem.

Awards and Achievements (continued)

- 08/24 ◇ **Womanium Quantum + AI 2024** [🔗](#) : Program finalist and QSL fellowship nominee.
- 06/24 ◇ **IBM Quantum Challenge 2024** [🔗](#) [🔗](#) : Ranked 11th globally.
- 01/16 ◇ **6th BAS Science Olympiad** [🔗](#) : Division 3rd and National level nominee.
- 09/15 ◇ **5th DSA Physics Olympiad** [🔗](#) : 7th Place.

Volunteering Work

- 04/25 – Present ◇ **Coordinator & Mentor, QBangladesh (QWorld)** [🔗](#) : Mentoring students and promoting quantum computing literacy in Bangladesh.
Hosted Events: QBronze [🔗](#) , QSilver [🔗](#) .

Hobbies

- ◇ Chess [🔗](#)
- ◇ Gardening

References

Dr. Md. Abdulla Al Mamun

Professor

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Hajee Mohammad Danesh Science and Technology University (HSTU)

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Dr. Masud Ibn Afjal

Professor

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